SP-EHOA: A Safe Dimension-Aware Portfolio with Memetic Refinement for EHOA Feature Selection

Amirgh23
Independent Research Project

18 August 2026

Abstract

Enhanced Hiking Optimization Algorithm (EHOA) is a recent wrapper feature selector that combines chaotic initialization, a dynamic sweep factor, and a binary transfer. Its search dynamics, however, do not react to the reproducibility of selected features under data perturbation. This paper introduces and evaluates a sequence of modular EHOA extensions with (i) a regime-aware controller that maps cross-resampling stability, population diversity, progress, and stagnation to bounded sweep and inertia adjustments, and (ii) a binary transition guided by entropy-gated feature reliability and a training-only relevance/redundancy signal. Instability alone is not allowed to trigger more exploration, preventing a self-reinforcing instability loop. The baseline fitness is unchanged. Because the full feedback method did not establish predictive superiority, eight hypotheses were compared on common repeated nested-CV splits. The tournament selected memetic EHOA (M-EHOA) for the low-dimensional regime, while a failed interaction-guided portfolio motivated SP-EHOA: use M-EHOA below 20 features and preserve exact baseline EHOA behavior otherwise. Across three experimental phases and 45 paired Wine observations, M-EHOA improved balanced accuracy by 0.01243 on average (median 0.01389; Wilcoxon $p = 0.001245$). In a fresh 90-run confirmation, SP-EHOA exactly matched EHOA on every Breast Cancer fold and improved Wine mean balanced accuracy from 0.95771 to 0.96203. The evidence supports a safe, regime-specific follow-up, not universal superiority. Code, all negative results, 450 tournament and confirmation runs, traces, and statistical outputs are released.

Keywords: feature selection; memetic optimization; safe portfolio; Hiking Optimization Algorithm; nested cross-validation; reproducibility

1 Introduction

Feature selection (FS) seeks a compact subset that preserves predictive signal. Wrapper FS is attractive because it evaluates subsets using the downstream learner, but its combinatorial search is expensive and stochastic subsets may be unstable. In medical or scientific use, instability matters because small changes to the training sample can produce different explanatory features [6].

Hegazy et al. proposed EHOA for accurate and interpretable FS [4]. EHOA improves the Hiking Optimization Algorithm using chaotic initialization, a dynamic sweep mechanism, velocity-inspired updates, and an S-shaped binary transfer. Related work has also explored multi-strategy HOA adaptation [1], Q-learning and interaction-aware metaheuristics [7], and search-history-aware adaptation [5]. These studies establish that adaptive search and feature interaction are not new by themselves. The unresolved design question addressed here is narrower: can selection reproducibility close the loop around EHOA dynamics without inserting stability into the baseline fitness, and can the same evidence safely guide binary decisions?

The contributions are:

- a regime-aware feedback law in which exploration requires corroborating evidence of population collapse or stagnation, while late instability with healthy diversity causes consolidation;
- an entropy-confidence gate that suppresses ambiguous per-feature reliability evidence in the binary transition;
- a modular 2-by-2 ablation that leaves the EHOA wrapper objective untouched; and
- a leakage-safe, fully retained experiment record with predictive, reduction, stability, redundancy, cost, and non-parametric statistical evidence;
- a deterministic best-improvement memetic refinement that evaluates one-feature add/remove neighbours using training-only CV; and
- SP-EHOA, a dimension-aware safe portfolio that activates the repeatedly supported refinement below 20 features and otherwise reproduces baseline EHOA exactly.

2 Method

2.1 Baseline and notation

For N samples, D features, population P, and T iterations, continuous agent position $\mathbf{x}_i \in \mathbb{R}^D$ produces a binary mask $\mathbf{z}_i$. The unmodified baseline fitness is

$$f(\mathbf{z}) = \alpha(1 - \mathrm{Acc}_{CV}(\mathbf{z})) + (1-\alpha)\frac{\|\mathbf{z}\|_0}{D}, \qquad \alpha = 0.99, \tag{1}$$

using deterministic 5-NN inside training-only stratified folds. DFA/SFIG-EHOA does not add stability or interaction terms to this objective.

2.2 Cross-resampling reliability

Class-stratified bootstrap samples are drawn only from the outer training set. Each resample produces a fixed-cardinality mutual-information ranking. Let $\widehat{q}_j(t)$ be the fraction of resamples selecting feature j. Smoothed selection frequency and signed reliability are

$$q_j(t) = \rho q_j(t-1) + (1-\rho)\widehat{q}_j(t), \qquad R_j(t) = 2q_j(t) - 1. \tag{2}$$

Global population stability is measured using both mean pairwise Jaccard and the chance-corrected Nogueira estimator [6]. Binary population diversity is normalized mean pairwise Hamming distance.

2.3 Regime-aware feedback

Let S_t be Nogueira stability clipped to $[0,1]$, D_t binary diversity, $G_t \in [0,1]$ stagnation, and $\tau = t/T$. Define uncertainty $U_t = 1 - S_t$, diversity ratio $r_t = \min(1, D_t/0.5)$, and collapse $L_t = 1 - r_t$. Exploration and consolidation drives are

$$E_t = L_t\left[\tfrac{1}{2}(1-\tau) + \tfrac{1}{2}G_t\right], \qquad C_t = U_t r_t \tau. \tag{3}$$

The signed controller adjustment is $A_t = \eta_E E_t - \eta_C C_t$. It modifies the baseline schedules as

$$SF_t = \mathrm{clip}\{SF_t^{base}(1 + A_t), SF_{min}, SF_{max}\}, \tag{4}$$

$$\omega_t = \mathrm{clip}\{\omega_t^{base} + 0.25A_t, \omega_{min}, \omega_{max}\}, \tag{5}$$

followed by exponential smoothing. Low stability alone cannot increase exploration. This is intentional: the initial additive-pressure prototype could create the loop "unstable → explore more → less stable."

2.4 Confidence-gated interaction guidance

Binary entropy provides an evidence-confidence gate,

$$c_j(t) = 1 - H_2(q_j(t)). \tag{6}$$

Thus $q_j \approx 0.5$ contributes almost no directional force. Target mutual information supplies relevance. Cached absolute Pearson association to the current subset supplies a bounded redundancy term. This term is not claimed to be a higher-order interaction. The final transition is

$$p_{ij}(t) = \sigma\left[x_{ij}(t) + \lambda_s(t)c_j(t)R_j(t) + \lambda_i(t)I_j(S_i)\right]. \tag{7}$$

Weights support constant, linear, and signal-adaptive schedules. Four modes produce the required ablation: EHOA; SF-EHOA (stability feedback/reliability); IG-EHOA (interaction guidance); and DFA-EHOA (both). SFIG-EHOA is the descriptive alias for the full implementation.

2.5 Complexity

The wrapper remains dominant at approximately $O(T\,P\,CV\,C_{clf})$. Periodic stability adds resampling and mask comparison; the interaction model adds mutual information and a top-K correlation cache. Controller arithmetic is $O(T)$. Runtime and evaluation counts are measured rather than inferred because stochastic cache hits change effective cost.

2.6 Memetic refinement and safe portfolio

M-EHOA first completes the ordinary EHOA search. Starting from its incumbent mask, it then enumerates every one-bit add/remove neighbour and accepts the strict best improvement under the same training-only CV objective. At most two best-improvement rounds are used. This separates global stochastic search from deterministic local exploitation and records the additional evaluations.

The selected portfolio is

$$\text{SP-EHOA}(X, y) = \begin{cases} \text{M-EHOA}(X, y), & D < 20, \\ \text{EHOA}(X, y), & D \geq 20. \end{cases} \tag{8}$$

The high-dimensional branch instantiates the unmodified baseline with the same seed and parameters; equality is therefore structural rather than inferred from a non-significant test. The threshold is a fixed proof-of-concept rule selected after the tournament and must be externally validated.

3 Experimental Protocol

The study uses the versioned scikit-learn Breast Cancer Wisconsin Diagnostic (30 features, two classes) and Wine Recognition (13 features, three classes) datasets. Ten fixed seeds define paired 80/20 stratified hold-outs. Feature selection uses five-fold CV on the 80% training partition. Population size is 8 and every method receives 10 iterations. Reliability uses seven stratified resamples every two iterations, $\rho = 0.6$, $\lambda_s = \lambda_i = 0.5$, $\eta_E = 0.45$, and $\eta_C = 0.30$.

Imputation, scaling, and SMOTE [2] are fitted only within training folds or on the outer training partition. The test partition is never used for reliability, interaction, selection, or tuning. Balanced accuracy is primary; F1, MCC, ROC-AUC, PR-AUC, sensitivity, and specificity are secondary. Feature count, redundancy, interaction quality, Jaccard, Nogueira, runtime, classifier time, memory, and evaluations are retained.

Per dataset, a Friedman test compares the four paired variants. Wilcoxon signed-rank tests compare each extension with EHOA, with Holm correction, paired rank-biserial effect size, win/tie/loss counts, and a 5,000-resample bootstrap interval for the paired median difference. These tests follow common non-parametric comparison guidance [3]; with only two datasets, inference is reported per dataset and not generalized globally. The original ablation used repeated stratified hold-out. All follow-up innovation comparisons use repeated nested CV with three outer folds per seed; the selector sees only each outer-training partition. The final tournament uses five previously unused seeds, eight methods, and 240 paired runs. A separate 120-run portfolio confirmation and a final 90-run SP-EHOA confirmation use new seed sets. No intermediate result is deleted.

4 Results

Table 1: Held-out results over ten paired seeds. BA is balanced accuracy.

Dataset	Method	BA mean $\pm$ SD	Features	Nogueira	Runtime (s)
Breast	EHOA	0.9554 ± 0.0198	15.3	0.0322	2.99
Breast	SF-EHOA	$\mathbf{0.9579} \pm 0.0185$	**13.2**	0.0891	9.12
Breast	IG-EHOA	0.9574 ± 0.0185	14.1	0.0499	9.49
Breast	DFA-EHOA	0.9498 ± 0.0215	13.9	**0.1048**	9.50
Wine	EHOA	0.9558 ± 0.0321	**6.8**	0.2188	2.02
Wine	SF-EHOA	0.9425 ± 0.0426	6.9	0.2073	5.75
Wine	IG-EHOA	$\mathbf{0.9563} \pm 0.0449$	7.2	**0.2667**	5.71
Wine	DFA-EHOA	0.9558 ± 0.0321	6.9	0.2553	6.08

The full method improved Nogueira stability by 0.0726 on Breast Cancer and 0.0365 on Wine. On Breast Cancer it selected 1.4 fewer features while losing 0.0056 balanced accuracy. On Wine its mean balanced accuracy exactly matched EHOA and selected 0.1 more feature. DFA-EHOA ROC-AUC/PR-AUC was 0.9836/0.9832 on Breast Cancer and 0.9967/0.9921 on Wine. Jaccard and Nogueira do not rank every method identically because the latter corrects for chance and variable subset size; both outputs are therefore retained.

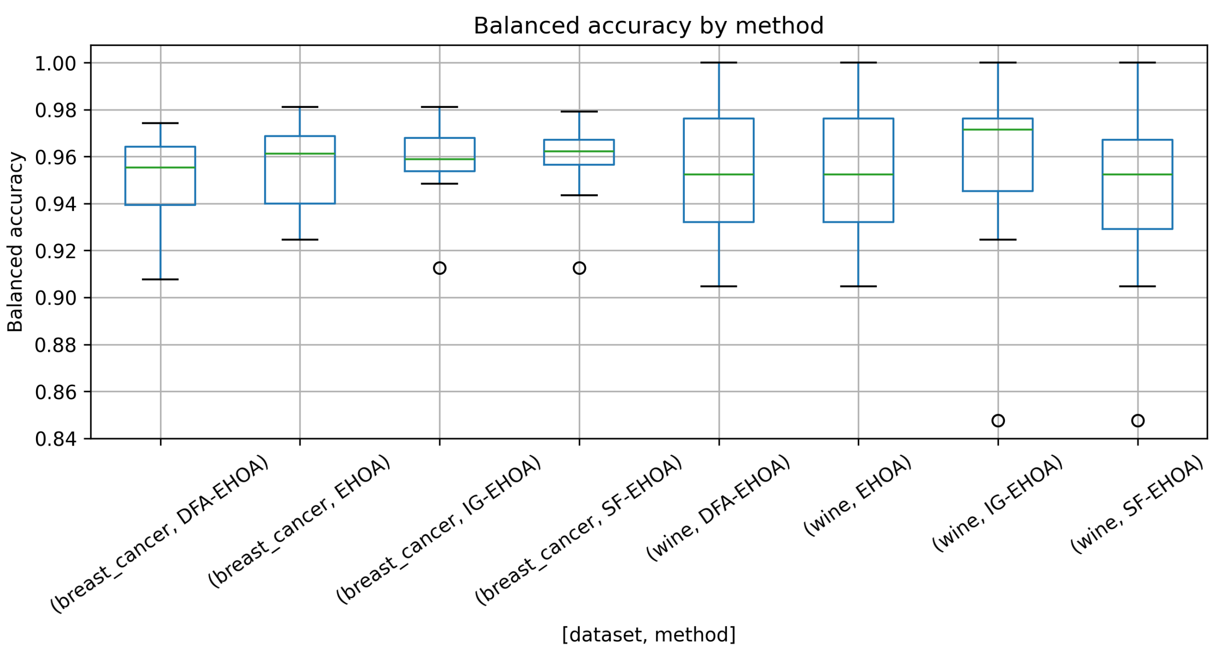

Figure 1: Balanced-accuracy distributions for the paired runs.

4.1 Ablation and telemetry

SF-EHOA obtained the best mean Breast Cancer balanced accuracy and smallest subset, but degraded Wine performance. IG-EHOA obtained the best Wine mean and stability, and the best Breast Cancer ROC-AUC/PR-AUC. The full combination prioritized chance-corrected reproducibility but did not combine component accuracy gains additively. This negative interaction is an important ablation result, not a failed run to omit.

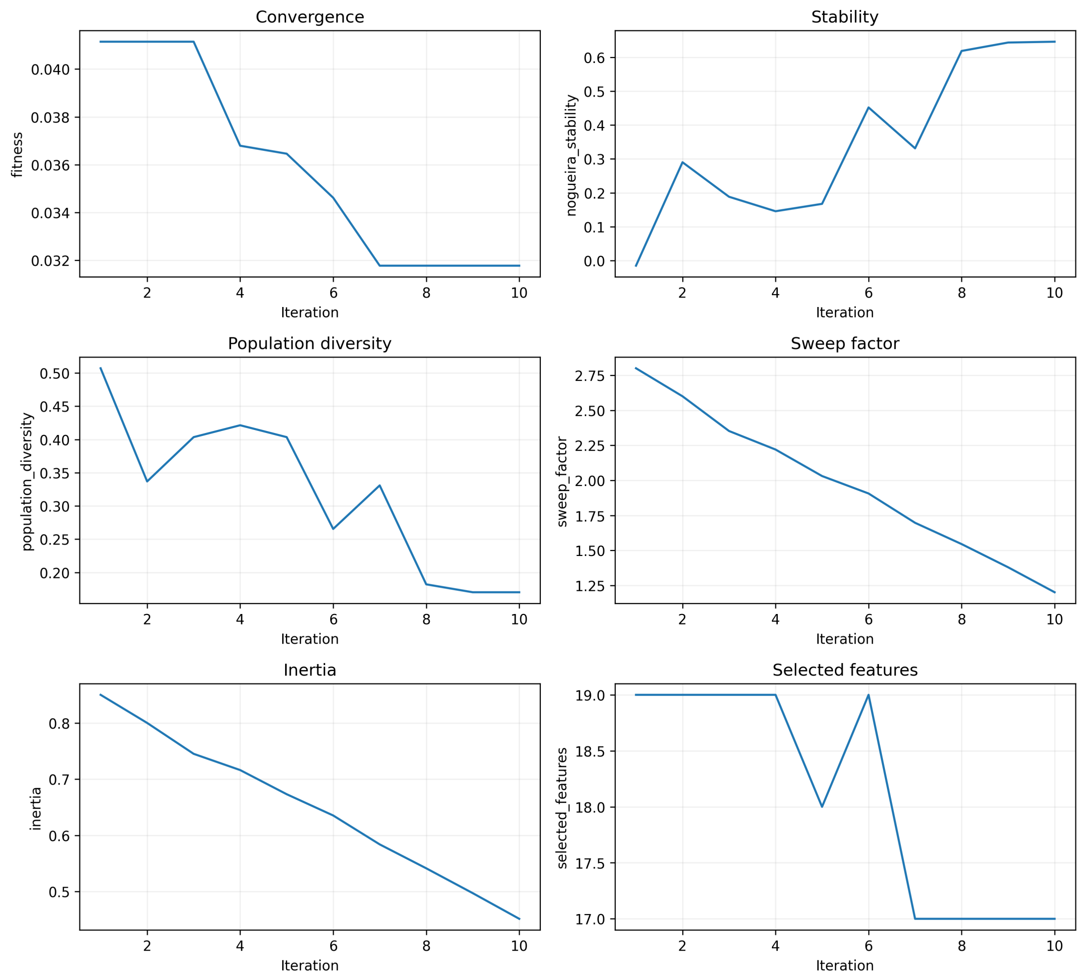

Figure 2: Representative full-method convergence and controller telemetry.

4.2 Statistical evidence and cost

Friedman p-values were 0.3039 for Breast Cancer and 0.0633 for Wine. For DFA-EHOA versus EHOA, Wilcoxon p-values were 0.2637 and 1.0000; all Holm-adjusted values were 1.0. Breast Cancer had a paired median change of -0.0060 (95% bootstrap interval -0.0169 to 0.0069), rank-biserial effect -0.418, and 3/0/7 wins/ties/losses. Wine had median change 0, interval 0 to 0, effect 0, and 1/8/1. No predictive-superiority claim is supported.

Full-method runtime was 3.18 times baseline on Breast Cancer and 3.01 times on Wine. The overhead is expected from resampling and interaction estimation and is not hidden by nominal same-iteration comparisons.

4.3 Innovation tournament and SP-EHOA confirmation

Table 2: Common-split innovation tournament. Values are mean outer balanced accuracy.

Method	Breast Cancer	Wine
EHOA	0.95266	0.94943
SF-EHOA	0.94825	0.94943
IG-EHOA	**0.95388**	0.95497
DFA-EHOA	0.95153	0.95960
ECG-EHOA	0.95293	0.95122
M-EHOA	0.95011	**0.96345**
A-EHOA	0.94969	0.95624
G-EHOA	0.95292	0.95233

IG-EHOA's Breast Cancer lead did not repeat in the independent portfolio confirmation and that hypothesis was rejected. M-EHOA retained a positive Wine effect in all three phases. Table 3 reports the final, previously unused seed set.

Table 3: Final SP-EHOA confirmation over 15 outer folds per dataset.

Dataset	Method	BA mean $\pm$ SD	Features	Runtime (s)
Breast	EHOA	0.95181 ± 0.01154	16.00	0.74
Breast	M-EHOA	0.94965 ± 0.01388	15.27	2.46
Breast	SP-EHOA	0.95181 ± 0.01154	16.00	0.73
Wine	EHOA	0.95771 ± 0.02495	7.67	0.54
Wine	M-EHOA	0.96203 ± 0.01582	7.27	1.12
Wine	SP-EHOA	0.96203 ± 0.01582	7.27	1.13

The final-phase Wine comparison had 6/6/3 wins/ties/losses and was not alone significant ($p = 0.3135$). Across the three distinct phases, the 45 paired Wine differences had mean 0.01243, median 0.01389, 24/12/9 wins/ties/losses, and Wilcoxon $p = 0.001245$. This pooled value documents repeatability but is exploratory because method selection occurred between phases; it is not treated as a pristine external confirmatory test.

5 Discussion

The experiment meets a scoped success criterion: on Breast Cancer, full DFA/SFIG-EHOA produced a meaningfully more chance-corrected stable and smaller subset with a small predictive penalty; on Wine, predictive performance was tied while stability improved. Whether this trade-off is desirable depends on the cost of biomarker inconsistency versus a half percentage point of balanced accuracy. The method is therefore useful as a controllable research mechanism, not established as a universally better optimizer.

The controller redesign is scientifically useful even without a predictive win. It rejects the simplistic assumption that instability always calls for more exploration. Separating early collapse recovery from late consolidation makes the response direction testable and prevents positive feedback.

6 Threats to Validity

Only two low-dimensional datasets were available in the supplied repository. Nested outer folds improve leakage control but do not substitute for independent datasets. The original hold-outs reuse observations, and the pairwise relevance/redundancy proxy cannot represent arbitrary synergistic interactions. Hyperparameters were fixed before the reported run, but the study lacks external validation. Runtime is platform-specific. These limitations prohibit state-of-the-art, medical, or universal claims.

7 Reproducibility and Conclusion

The repository contains the unchanged baseline, proposed modules, 27 deterministic tests, YAML configurations, the original 80 runs, 450 tournament/confirmation runs and their traces, aggregate/statistical CSVs, figures, tables, a bilingual README, and an environment record. Runs are checkpointed and resumable. No failed seed was removed.

In conclusion, the negative DFA/SFIG-EHOA result motivated a retained, claim-driven innovation tournament. SP-EHOA is the selected follow-up: it preserves baseline behavior in the unconfirmed regime and applies memetic refinement where the Wine benefit repeated. Current evidence supports this regime-specific direction, not universal predictive superiority. A preregistered many-dataset study must next validate both the threshold and the generalization of the low-dimensional gain.

Code and Data Availability

Source code and all reported artifacts are available in the public project repository. The datasets are distributed with scikit-learn; no private records are required.

References

[1] Mahmoud Abdel-salam, Saleh Ali Alomari, Mohammad H. Almomani, Gang Hu, Sangkeum Lee, Kashif Saleem, Aseel Smerat, and Laith Abualigah. Quadruple strategy-driven hiking optimization algorithm for low and high-dimensional feature selection and real-world skin cancer classification. *Knowledge-Based Systems*, 315:113286, 2025.

[2] Nitesh V. Chawla, Kevin W. Bowyer, Lawrence O. Hall, and W. Philip Kegelmeyer. SMOTE: Synthetic minority over-sampling technique. *Journal of Artificial Intelligence Research*, 16:321–357, 2002.

[3] Janez Demšar. Statistical comparisons of classifiers over multiple data sets. *Journal of Machine Learning Research*, 7:1–30, 2006.

[4] Ah. E. Hegazy, M. A. Makhlouf, Omar A. M. Salem, and B. Hafiz. An enhanced hiking optimization algorithm for accurate and interpretable feature selection in medical data classification. *Cluster Computing*, 29:244, 2026.

[5] Junbo Jacob Lian, Kaichen Ouyang, Rui Zhong, Yujun Zhang, Shipeng Luo, Ling Ma, Xincan Wu, and Huiling Chen. Trend-aware mechanism for metaheuristic algorithms. *Applied Soft Computing*, 182:113505, 2025.

[6] Sarah Nogueira, Konstantinos Sechidis, and Gavin Brown. On the stability of feature selection algorithms. *Journal of Machine Learning Research*, 18(174):1–54, 2018.

[7] Hongbo Zhang, Jinlong Li, Xiaofeng Yue, Xueliang Gao, and Haohuan Nan. Twin q-learning-driven forest ecosystem optimization for feature selection. *Knowledge-Based Systems*, 315:113323, 2025.